

MATHEMATICS MOST IMPORTANT TOPICS

Expert Revision Checklist & Exam Pattern Analysis

Created for Thunderstudy By Aashi

1. Algebra (Matrices & Determinants)

- **Solving Systems of Linear Equations:** Guaranteed 5-mark question using the matrix method ($AX = B$) or finding A^{-1} .
- **Matrix Properties:** Symmetric/Skew-symmetric matrices and properties of determinants for objective questions.

2. Calculus (Highest Weightage)

- **Applications of Integrals:** Area bounded by curves (circles, parabolas, ellipses) under integration.
- **Definite Integrals:** Focus on properties, especially the "King's Rule" [$f(x) = f(a+b-x)$].
- **Maxima and Minima:** Optimization word problems, frequently used in Case Studies.
- **Differential Equations:** Integrating Factor (I.F.) and Homogeneous equations.
- **Continuity:** Finding 'k' to make a function continuous.

3. Vectors & 3D Geometry

- **Shortest Distance:** Frequent 5-mark question for the distance between two skew lines.
- **Foot/Image:** Finding the foot of the perpendicular or image of a point across a line.
- **Vector Operations:** Dot/Cross product properties and area of parallelograms.

4. Probability & LPP

- **Bayes' Theorem:** High weightage conditional probability scenarios.
- **Probability Distribution:** Random variable distributions (e.g., coin tosses).
- **LPP Graphical Method:** Maximizing/Minimizing objective function Z subject to constraints.

5. Relations & Functions

- **Equivalence Relations:** Proving Reflexive, Symmetric, and Transitive properties.
- **Bijjective Functions:** Checking for One-One (Injective) and Onto (Surjective) mappings.

MASTER THESE 5-MARK "SURE-SHOT" QUESTIONS

1. Matrix Method for Linear Equations.
2. Shortest Distance between Skew Lines (3D Geometry).
3. Area Under the Curve (Integration).
4. Linear Programming (Graphical Method) OR Maxima/Minima Word Problems.

Mathematics Strategy By Aashi Thunderstudy
Analyze. Solve. Excel.